

Discussion of Bridging Structured Knowledge and Data: A Unified Framework with Finance Applications

(Cao, Chen, Cong & Shi: SKINNs)

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Discussion: Bridging Structured Knowledge & Data (SKINNs)

A flexible net f_θ and a differentiable theory g_ϕ , trained *together*:

$$\mathcal{L}(\theta, \phi) = \frac{1}{N} \sum_i (f_\theta(X_i) - y_i)^2 + \lambda \frac{1}{M} \sum_j (f_\theta(X_j^{\text{grid}}) - g_\phi(X_j^{\text{SK}}))^2.$$

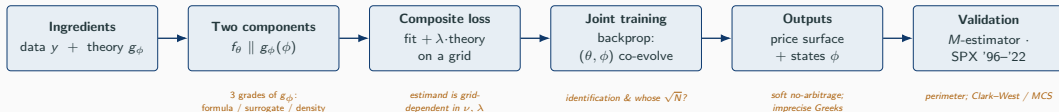
- Two moves: θ, ϕ learned *jointly*; $\phi =$ *interpretable* latent state (volatility, a risk-neutral density).
- g_ϕ in three grades: **parametric** (BS/Heston) $\rightarrow$ **surrogate** of a costly simulator $\rightarrow$ **non-parametric** density / autoencoder.

What the paper offers

Theory: cast as an M -estimator $\Rightarrow$ consistency, $\sqrt{N}$ normality, sandwich covariance. **Empirics:** on S&P 500 vanilla calls (1996–2022), compares favorably to NN, PINN, transfer learning, and structural models.

The article as one pipeline, then I attach my comments

Motivation: principled-but-rigid theory vs flexible-but-fragile data



Read left to right: the paper turns *data + theory* into *predictions + interpretable economics*, all in *one* optimisation. The amber notes mark where my comments attach, so the rest of the talk simply walks this line.

One object, several lenses, and how I read it

Strip the names and it is **regularized empirical-risk minimization** with a learned, moving target. The paper reads that object through three familiar lenses:

- **GMM** with weight $\text{diag}(1, \lambda)$, deliberately *not* the efficient weight; λ is a regularization dial, by design.
- **Bayesian MAP**: a Gaussian “theory prior” centered at g_ϕ , whose center is itself learned (empirical Bayes).
- **Variational**: minimize an energy, but the constraint is the *solution* g_ϕ , not a differential operator.

How I'll frame it

The diagnosis (rigid theory vs. fragile data) is right and the construction is elegant. I read these three as *illuminating reinterpretations* rather than strict nesting, e.g. it recovers MAP, not the full posterior; regularized, not efficient, GMM. $\Rightarrow$ two strengths, three points to discuss, one constructive variant.

What is genuinely new here, one of them deep

1. Constrain the solution, not the operator.

Why their Fig. 6 (PINN) is ill-conditioned and Fig. 7 (SKINN) is clean: a PINN differentiates the network, scaling its *neural-tangent-kernel* eigenvalues by powers of frequency $\Rightarrow$ ill-conditioning. SKINN matches a *zeroth-order* answer $f_\theta - g_\phi \Rightarrow$ the kernel is intact and both losses train together. Bonus: it admits models with *unobservable* states (Heston variance) that are out of reach for PINNs. This distinction is real and practically useful.

2. Joint learning as a learned-center prior, plus a clean taxonomy.

Theory and net co-evolve (empirical Bayes by backprop) rather than freeze-then-fit, and the parametric $\rightarrow$ surrogate $\rightarrow$ non-parametric ladder is a genuinely useful organizing device. The most original instance is the *non-parametric no-arbitrage* variant: learning a high-dimensional risk-neutral density directly from prices.

Where it sits: generous neighbors worth citing

The core pattern (*embed a differentiable theory module, learn its parameters jointly*) has close cousins. Naming them would sharpen the contribution rather than dilute it:

- **Inverse PINNs** (Raissi et al. 2019) already co-estimate model parameters; the real delta here is solution- vs. operator-matching.
- **Differentiable calibration / neural-SDE** (Horvath–Muguruza–Tomas 2021; Cuchiero–Khosrawi–Teichmann 2020) $\approx$ the semi-parametric (surrogate) SKINN.
- **No-arbitrage in a neural SDF** (Chen–Pelger–Zhu 2024) is close kin to the non-parametric variant.
- **Risk-neutral density estimation** (Breedon–Litzenberger 1978; Jackwerth 2000; Fengler 2009) is where the 2,000-dim $\mathbb{Q}$ connects.
- **Universal differential equations** (Rackauckas 2020); **informed-ML taxonomies** (von Rueden 2021); **shrinkage to a parametric start** (Hjort–Glad 1995).

The distinctive combination: solution-matching, over a collocation domain, with structural ϕ learned jointly, shown at scale, including a non-parametric no-arbitrage form.

Point 1: sharpening the $\sqrt{N}$ statement

Theorems 1–2 assume a *unique* interior minimizer and an *invertible* Hessian. For an over-parameterized net these are hard to sustain:

permutation & ReLU-rescaling symmetry $\Rightarrow \nabla^2 \mathcal{L}$ singular, θ^* not a point.

Also, with the architecture fixed, this is *finite-dimensional* M -estimation, so $\sqrt{N}$ is expected, not surprising; the “despite infinite dimensions” framing can be eased.

A fix that strengthens the paper

Don't claim normality for θ . Treat f_θ as an *infinite-dimensional nuisance*; profile it out, $\theta(\phi)$; use a **Neyman-orthogonal** score for the low-dim ϕ . This is debiased / double-ML, the natural vehicle for inference on the interpretable parameters, and $\sqrt{N}$ survives for ϕ .

Relatedly, ϕ is identified only if $\text{rank } \mathbb{E}_\nu[\partial_\phi g_\phi] = \dim \phi$; a flat g_ϕ behaves like a *weak instrument* (Wald intervals under-cover).

Point 2: what ϕ estimates, and soft no-arbitrage

- The theory term integrates over a *collocation measure* ν the analyst picks $\Rightarrow \phi^* = \phi^*(\nu, \lambda)$ is a regularized (pseudo-true) object, not necessarily a market primitive. **Declare ν ; show stability.**
- No-arbitrage lives in g_ϕ , but prices come from f_θ , tied only by a soft $\lambda \Rightarrow$ no-arbitrage enters only softly, so reported prices need not be arbitrage-free.
- Suggestive caveat on interpretation: even the *theory-free* autoencoder factors predict next-month VIX, so “ ϕ is economically meaningful” is partly mechanical (prices already encode vol). Worth disentangling.
- The density model pins only the per-maturity *marginal* (Breedon–Litzenberger); the joint law is free, and piecewise-constant-in- τ may not satisfy **convex order** (calendar arbitrage).

Remedy: price from g_ϕ (or project onto the no-arbitrage cone) and impose convex order, making the consistency *structural*, not just penalized.

Point 3: the test bed, hedging, and scoring

Test bed: European S&P 500 *vanilla calls*, one index, 7–365 days, the most favorable corner of the framework’s range.

- Add an **arbitrage-free surface** baseline (SVI/SSVI/GP), the desk standard; report **implied-vol** / vega-weighted error, since price-RMSE is dominated by intrinsic value.
- **Pricing-optimal** $\neq$ **hedging-optimal**: the Delta is autodiff of a *level*-fitted net. Put Greeks in the loss, $\|\partial_m f_\theta - \partial_m g_\phi\|^2$ (**Sobolev training**), or go deep-hedging.
- Inference: nested pairs $\rightarrow$ **Clark–West**; hundreds of comparisons $\rightarrow$ **Model Confidence Set**.

A frontier data point (Guyon)

SPX prices carry little forward-vol information; neural objects fit under the right hard constraints (joint SPX/VIX) tend to recover *path-dependent, mean-reverting* volatility. A Markovian “interpretable state” is a reasonable first pass, but recent evidence points elsewhere, a natural robustness check.

(Minor: AE composition order; θ overloaded; uniform-vs-LHS sampling; duplicated table columns; twin captions; broken Appendix ?? ref; a 9-dim surrogate on 5×10^7 pts, report held-out error.)

A constructive “SKINN 2.0”, and where I land

Keep the framework; refine the instantiation:

1. g_ϕ = **path-dependent volatility** (Guyon–Lekeufack) alongside Markovian Heston, as a robustness ladder.
2. Add **VIX** futures & smiles, with the log-contract identity as a **hard** constraint (couples the markets).
3. **Sobolev** / **deep-hedging** loss; **orthogonalized** inference for ϕ ; declare ν .
4. Benchmark vs. an arbitrage-free surface; score by **MCS**, in IV-space, net of costs.

Takeaway

An elegant idea on the right problem. As written, a well-grounded *statistical estimation* framework; a few additions (structural pricing, orthogonal inference, a stronger test bed) would turn it into a competitive *pricing engine*. The fixes are clear, principled, and within reach. I learned a lot from it.